

Digital Recipe Book

Final Project Report

Interactive Web Application Development

I Year | II Semester

ICT 1209 - Web Technologies

Department of ICT

Faculty of Technology

Rajarata University of Sri Lanka

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1. Project Overview

1.1 Purpose of the Project

Cooking is an essential daily activity and an art that connects cultures, families, and traditions. However, many home cooks, students, and food enthusiasts struggle with disorganized culinary recipes scattered across social media, random websites, and physical notebooks. Finding reliable cooking steps, accurate ingredient measurements, and meal ideas often becomes a time-consuming and frustrating task.

To solve this problem, we created "**Digital Recipe Book**" as our mini project for ICT 1209 (Web Technologies). Our main goal is to build an easy-to-use, responsive, and database-driven web platform where people can discover different delicious food ideas, sort recipes by meal categories (Breakfast, Lunch, Dinner, Dessert, Drinks, Snacks), search in real-time, view detailed step-by-step cooking instructions, and easily submit their own favorite dishes to share with others.

1.2 Target Audience

Our web application is built for people who want a simple way to explore and organize culinary recipes:

- **Home Cooks:** People who prepare daily family meals and want an organized collection of tasty dishes.
- **Students and Busy Professionals:** Busy individuals who need quick and simple 15–20 minute breakfast, lunch, or snack recipes with clear instructions.
- **Food Lovers & Culture Enthusiasts:** Anyone wanting to explore authentic culinary dishes alongside popular global meals.
- **Community Recipe Creators:** Users who enjoy cooking and want to share their own culinary creations with preparation times, ingredients, and instructions.

1.3 Key Features

The core features of our website include:

- **User Registration and Login:** New users can create an account with a username, email, and password. Registered users can log in securely and log out anytime.
- **Interactive Global Navigation Bar:** A dynamic navbar that automatically adapts based on session authentication state (displays User Profile and Logout when logged in, or Login when logged out).
- **Recipe Catalog and Filtering:** A dedicated recipes page where users can browse dishes and filter them by category (All, Breakfast, Lunch, Dinner, Dessert, Drinks, Snacks) using JavaScript.
- **Live Real-Time Recipe Search:** A keyboard-accessible instant search popup (Ctrl + K) that filters recipes in real time as the user types without refreshing the page.

- **Interactive Recipe Details Modal:** Clicking any recipe card opens an interactive popup modal displaying cooking time, user rating, ingredients lists, and instructions.
- **User Dashboard for Recipe Uploads:** A private dashboard where logged-in users can fill out a form to submit new recipes directly to the database.
- **Automatic Recipe Display:** Any recipe submitted by a user gets saved to the MySQL database and immediately appears on the recipes catalog page.
- **Contact Us Form:** A page where visitors can send questions or feedback, which gets stored in the database.

1.4 Technology Stack

The project was developed using the required technologies for ICT 1209:

Layer	Technology	Description
Frontend Structure	HTML5 and Bootstrap 5	Responsive grid layout, navbar, cards, forms, and modal structures.
Styling	CSS3	Custom styling, animations, responsive media queries, and sticky footer.
Client-Side Scripting	JavaScript (Vanilla ES6+)	Category filtering buttons, real-time live search (Ctrl + K), and modal popups.
Backend Scripting	PHP 8 (PDO)	Session handling, secure user login with BCrypt, form processing, and database CRUD.
Database	MySQL (XAMPP)	Relational database storing users, submitted recipes, and contact messages.
Version Control	Git and GitHub	Source code version control, branch management, and collaborative commit tracking.

2. System Design

2.1 Database Schema and Tables

The application uses a MySQL database named `recipe_db`. It contains three related tables to handle users, their submitted recipes, and contact inquiries:

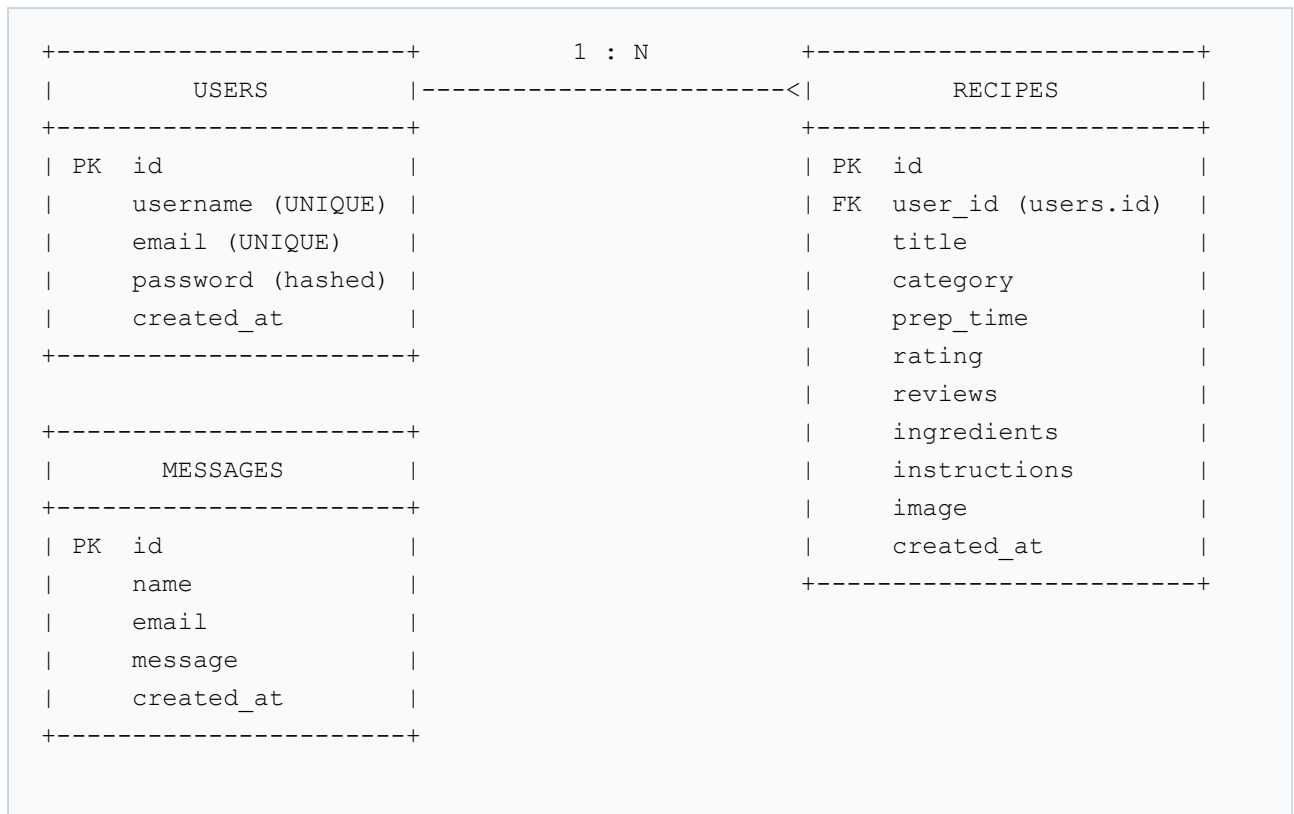


Figure 1: Entity-Relationship Diagram (ERD) of recipe_db Database

Database Relationship: One user can submit multiple recipes (1 : N). The `user_id` field in the `recipes` table connects each recipe to the user who uploaded it (`ON DELETE SET NULL`).

2.2 Database Table Specifications

Table	Field Name	Data Type	Key / Null	Description
users	id	INT	PK (Auto-Inc)	Unique User ID
users	username	VARCHAR(50)	NOT NULL	User display name / username
users	email	VARCHAR(100)	UNIQUE	User login email
users	password	VARCHAR(255)	NOT NULL	BCRYPT hashed password
users	created_at	TIMESTAMP	CURRENT_TIMESTAMP	Registration timestamp
recipes	id	INT	PK (Auto-Inc)	Unique Recipe ID
recipes	user_id	INT	FK (users.id)	ID of recipe author (Nullable)
recipes	title	VARCHAR(150)	NOT NULL	Title of the recipe
recipes	category	VARCHAR(50)	NOT NULL	Meal category (Breakfast, Lunch, etc.)
recipes	prep_time	VARCHAR(30)	NOT NULL	Estimated cooking duration
recipes	rating	DECIMAL(3,1)	DEFAULT 4.5	User rating score
recipes	ingredients	TEXT	NOT NULL	Required ingredients list
recipes	instructions	TEXT	NOT NULL	Preparation steps
recipes	image	VARCHAR(255)	DEFAULT ...	Path to recipe image
recipes	created_at	TIMESTAMP	CURRENT_TIMESTAMP	Submission timestamp
messages	id	INT	PK (Auto-Inc)	Unique Message ID

Table	Field Name	Data Type	Key / Null	Description
messages	name	VARCHAR(100)	NOT NULL	Sender full name
messages	email	VARCHAR(100)	NOT NULL	Sender email address
messages	message	TEXT	NOT NULL	Message content
messages	created_at	TIMESTAMP	CURRENT_TIMESTAMP	Submission timestamp

2.3 Folder Structure

Our project folder is organized cleanly as required by the course guidelines:

```

digital-recipe-book/
├── auth/                                # Login, Register and Logout PHP scripts
│   ├── login.php
│   ├── logout.php
│   └── register.php
├── assets/                              # Styling and images folder
│   ├── css/style.css
│   └── images/
├── includes/                            # Core backend PHP includes
│   ├── db.php                          # PDO database connection
│   └── functions.php                    # Session, sanitize and auth helper functions
├── about.php                            # About Us page
├── app.js                               # Client-side JavaScript (search modal & filter)
├── contact.php                          # Contact form with database insertion
├── dashboard.php                        # User recipe submission dashboard
├── database.sql                         # Database schema export file
├── index.php                            # Home page with search bar & popular recipes
├── recipes.php                          # Recipe catalog with dynamic database cards
├── styles.css                           # Custom responsive stylesheet
└── README.md                            # Project description and setup guide

```

3. Implementation

This section describes the key features of the application along with screenshots showing the user interface.

3.1 Home Page and Live Search Modal (`index.php`)

The home page introduces visitors to Digital Recipe Book. It features a hero banner with instant search, category chips, popular recipes grid, and an interactive live search popup modal accessible via the search icon or `Ctrl + K`.



Figure 2: Home Page and Popular Recipes Grid

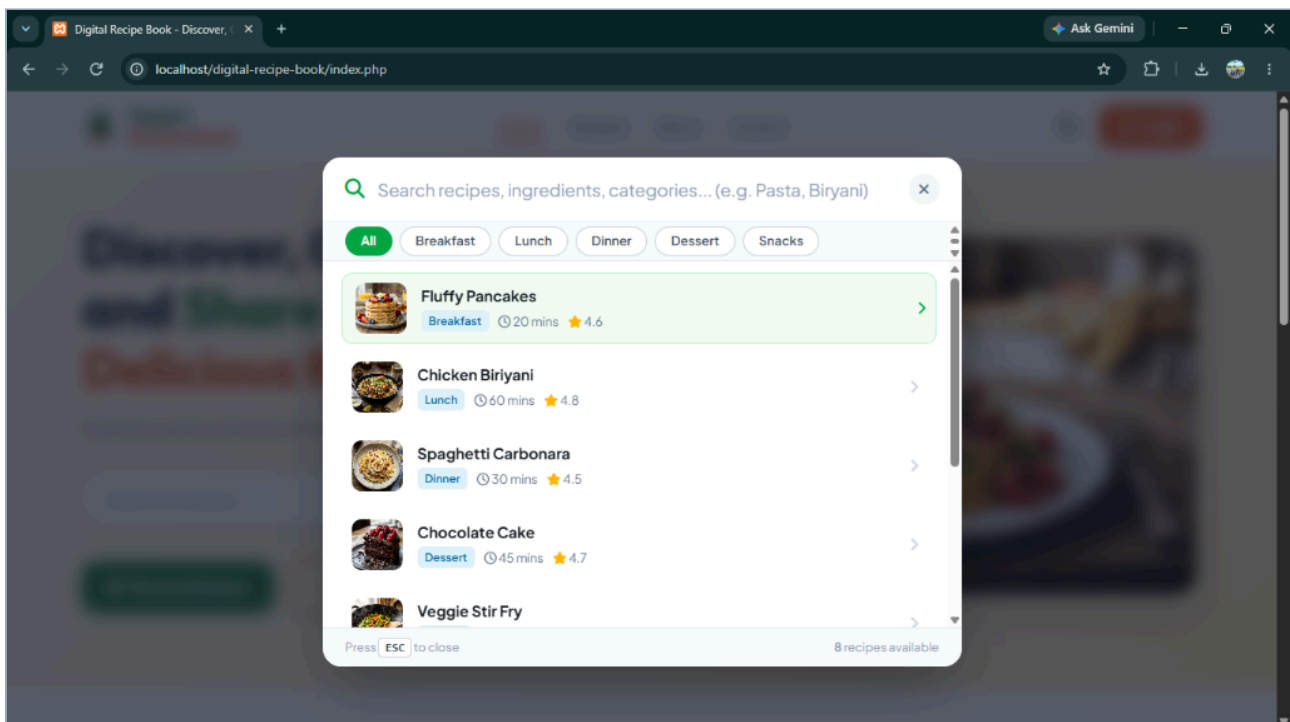


Figure 3: Live Real-Time Search Modal Interface (`Ctrl + K`)

3.2 Recipe Catalog with Filtering and Search (`recipes.php`)

`recipes.php` displays all available recipes in a 4-column card layout queried directly from the MySQL database. Users can filter recipes by clicking category buttons on the sidebar or top dropdown. Clicking any recipe card opens an interactive popup modal displaying cooking times, ingredients, and instructions.

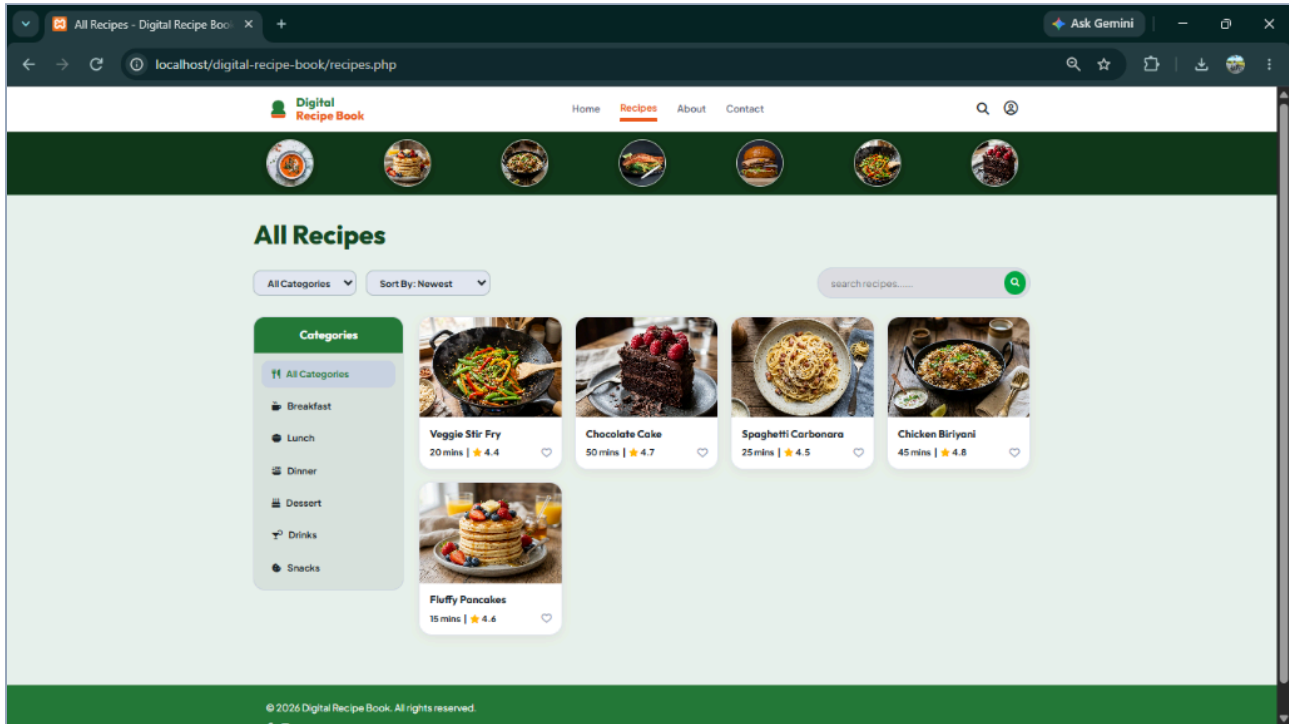


Figure 4: Recipe Catalog with Category Sidebar Filters

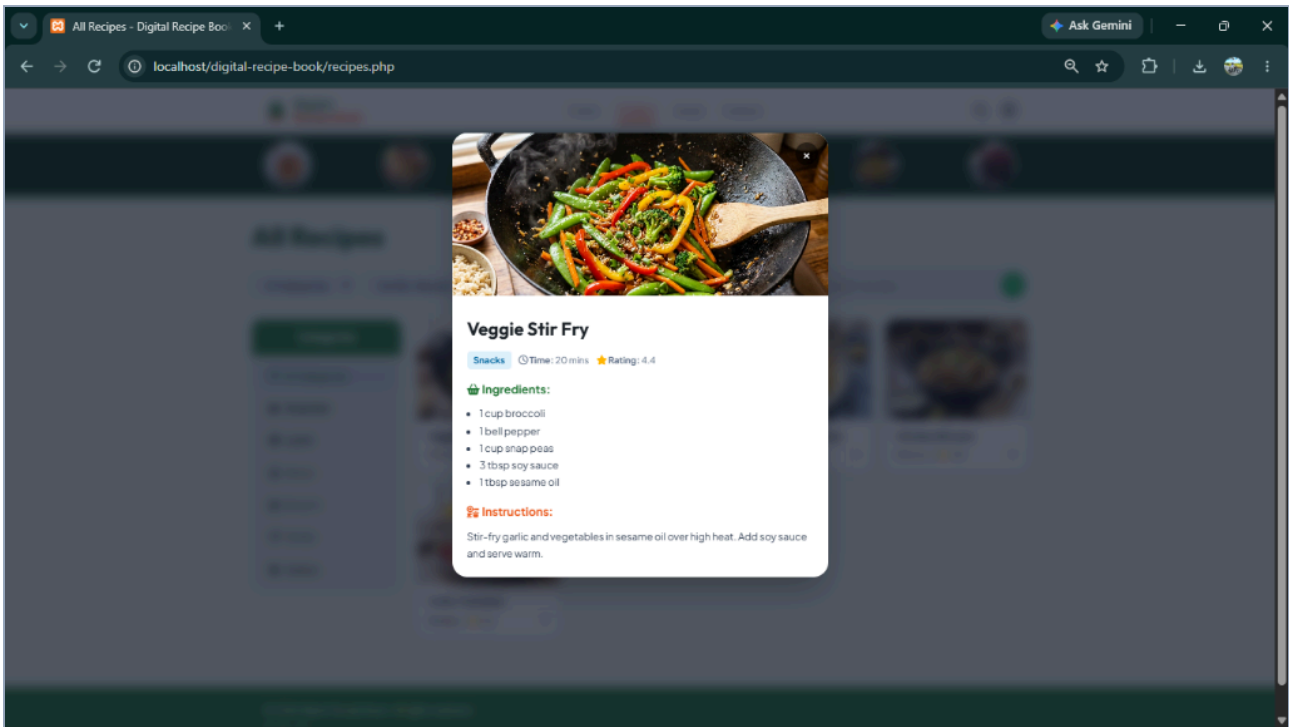


Figure 5: Interactive Recipe Detail Popup Modal

3.3 User Authentication (`register.php` and `login.php`)

User registration and login are handled securely using PHP and MySQL. Passwords are encrypted using `password_hash()` with BCRYPT before saving to the database. When logging in, credentials are verified using `password_verify()`. After successful login, `session_regenerate_id()` is called for session security and the user is redirected to their Dashboard. The navigation bar dynamically updates to show the user's username and Logout button.

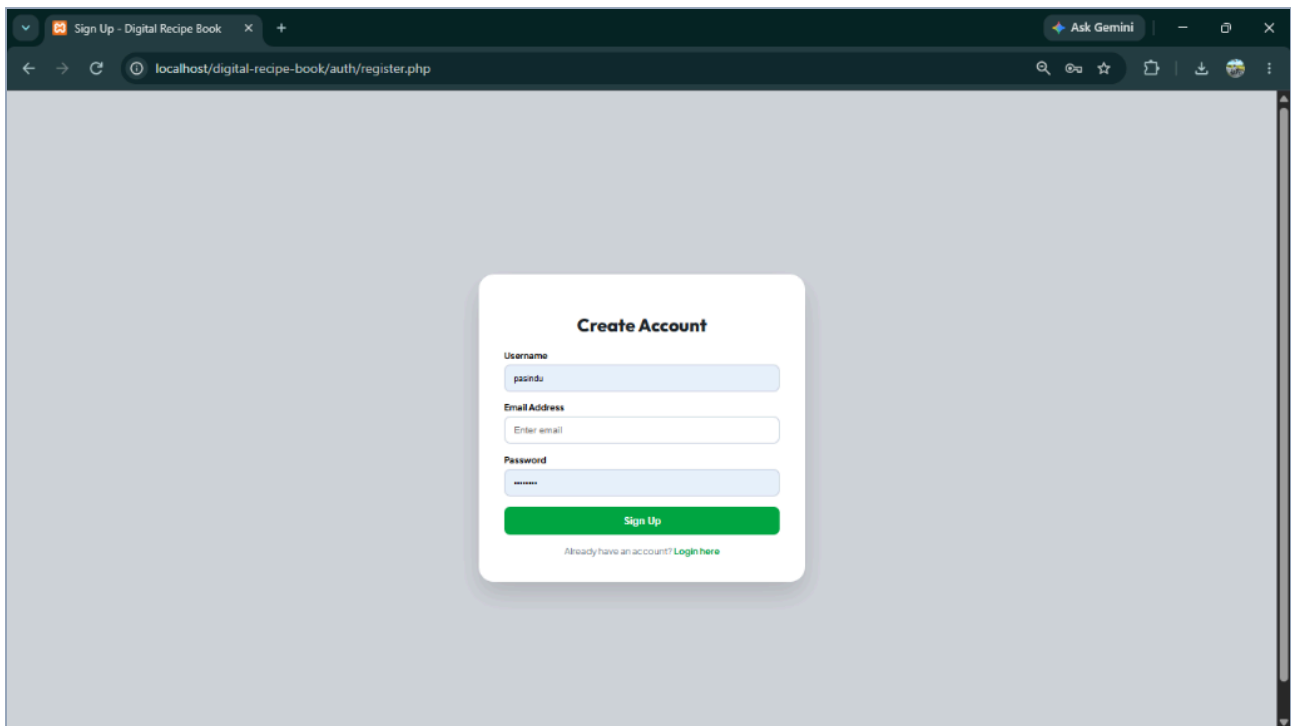


Figure 6: User Registration Page

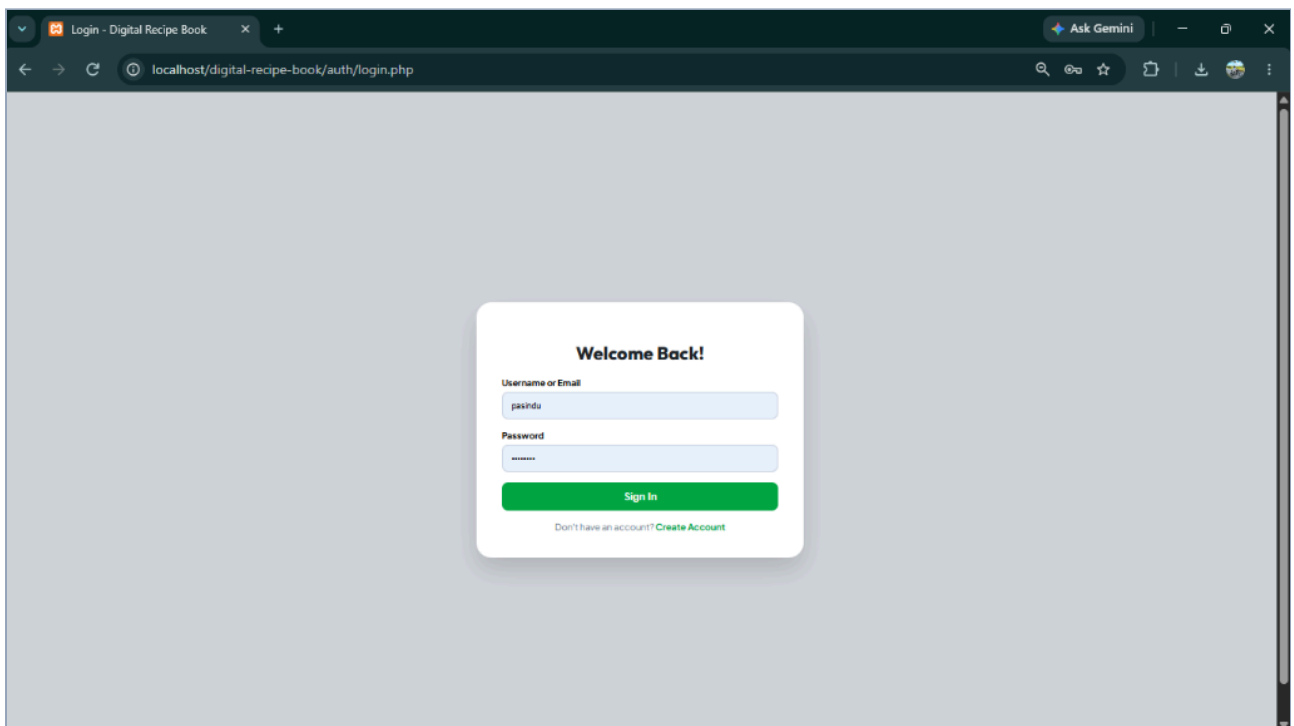


Figure 7: User Login Page and Dynamic Navbar

3.4 User Recipe Submission Dashboard (`dashboard.php`)

Only logged-in users can access `dashboard.php` . If an unauthenticated guest tries to visit this page, they are automatically redirected to `auth/login.php` . The dashboard contains a clean form where

users enter recipe title, select category, preparation time, list ingredients, and write cooking instructions to publish directly to MySQL.

The screenshot displays a web browser window with the URL `localhost/digital-recipe-book/dashboard.php`. The page features a navigation bar with links for Home, Recipes, and Contact, and a red Logout button. The main content area is divided into two columns. The left column, titled 'Add New Recipe', contains a form with the following fields: 'Recipe Title' (with a placeholder 'e.g. Garlic Butter Shrimp'), 'Category' (a dropdown menu currently showing 'Lunch'), 'Preparation Time' (with a placeholder 'e.g. 25 mins'), 'Ingredients (comma separated)' (with a placeholder 'Shrimp, butter, garlic, lemon juice, parsley'), and 'Cooking Instructions' (with a placeholder 'Melt butter, saute garlic, cook shrimp for 4 minutes and garnish.'). A green 'Publish to Database' button is at the bottom of this form. The right column, titled 'My Submitted Recipes (0)', shows a message: 'You haven't submitted any recipes yet. Use the form to publish your first recipe!'.

Figure 8: User Dashboard and Recipe Submission Form

3.5 Contact Us Form and About Us Page (`contact.php` and `about.php`)

`contact.php` allows visitors to send feedback or inquiries. The form collects Name, Email, and Message, validates the fields, and stores the message in the `messages` table. A green confirmation banner notifies the user of success. `about.php` presents the culinary mission and platform statistics.

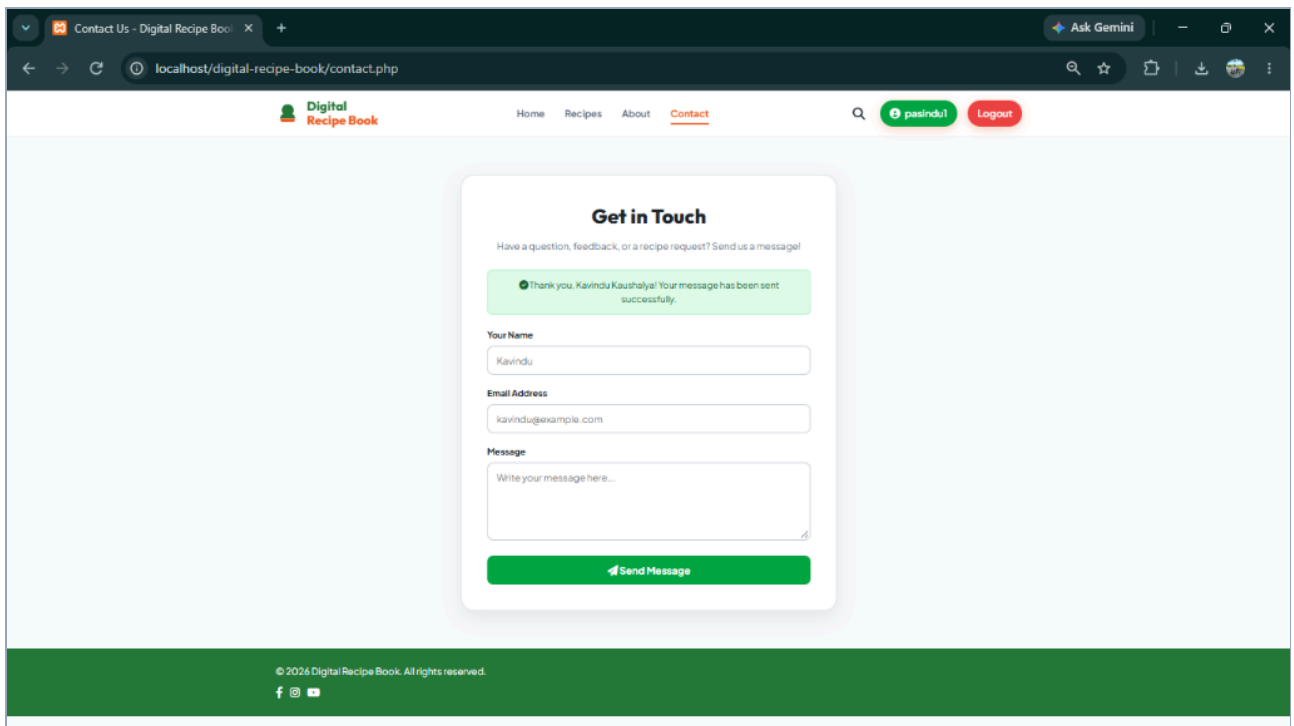


Figure 9: Contact Us Form with Success Alert

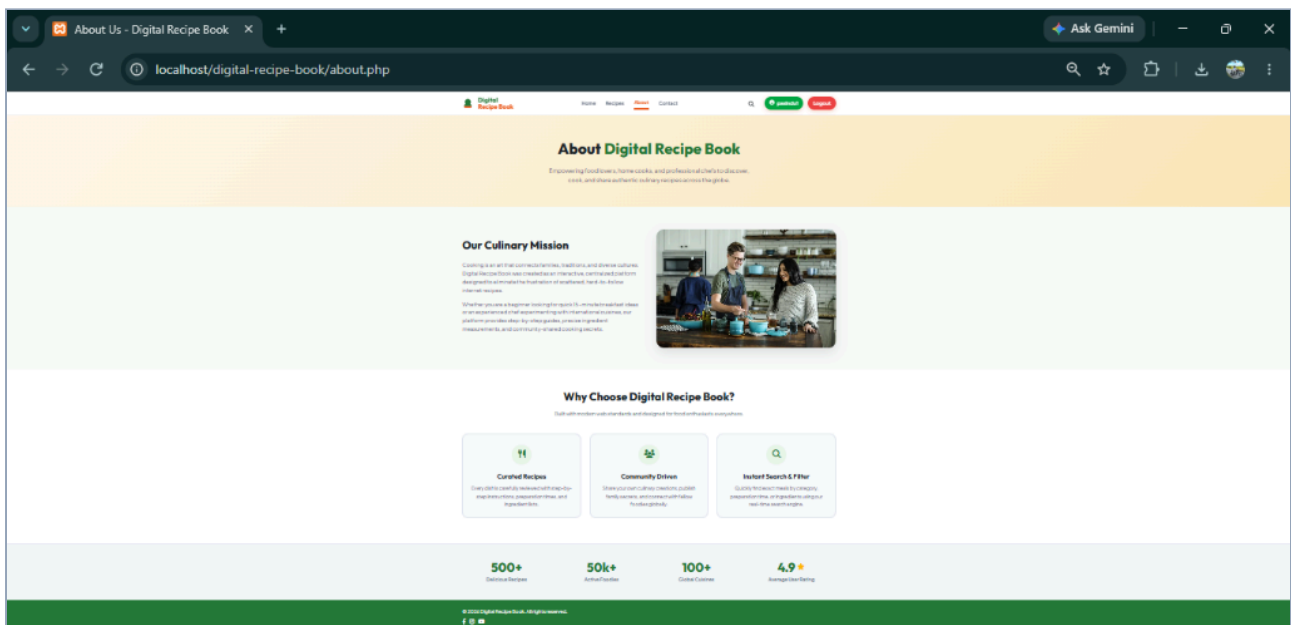


Figure 10: About Us Page and Statistics Banner

3.6 Database Verification in phpMyAdmin

All user accounts, submitted recipes, and inquiries are stored and managed in the MySQL `recipe_db` database via phpMyAdmin.

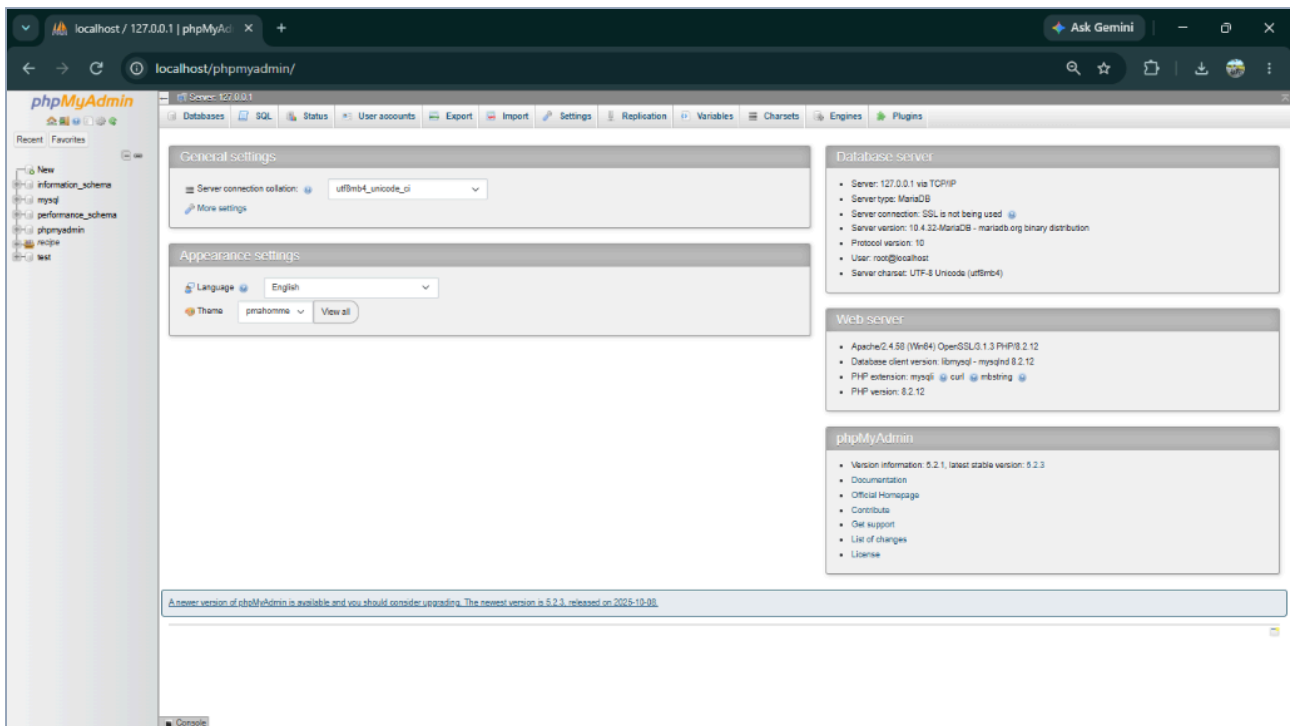


Figure 11: phpMyAdmin Database Schema and Tables

4. Challenges and Solutions

During the development of the project, we encountered several practical challenges and solved them as follows:

4.1 SQL Injection Vulnerabilities & Input Sanitization

- **Problem:** Early SQL queries concatenated raw user inputs directly into query strings, creating critical security risks.
- **Solution:** Enforced **PDO Prepared Statements** (`$stmt = $pdo->prepare(...)`) with parameter binding across all database transactions, and sanitized visitor input with `htmlspecialchars()` and `stripslashes()`.

4.2 Dynamic Modal Popup Integration in Catalog

- **Problem:** When transitioning from static recipe cards to dynamic MySQL queries in `recipes.php`, clicking recipe cards failed to open detail popups because the query omitted ingredients and instructions fields.
- **Solution:** Updated the backend PDO query to fetch `ingredients` and `instructions`, created a modal popup container in `recipes.php`, and attached dynamic JavaScript click event listeners to populate the modal on demand.

4.3 External Image Fallback Architecture

- **Problem:** If local image files were missing or remote image URLs became unavailable, recipe cards displayed broken image placeholders.
- **Solution:** Implemented a dual-layer image fallback mechanism utilizing JavaScript `onerror` event handlers (`onerror="this.onerror=null; this.src='assets/images/hero_pasta_dish.jpg';"`) guaranteeing high-resolution visuals at all times.

4.4 Multi-Device Responsive Grid Alignment

- **Problem:** Horizontal category filter chips and recipe card grids overflowed horizontally on smaller mobile viewports.
- **Solution:** Implemented responsive CSS Grid layouts with `repeat(auto-fit, minmax(260px, 1fr))` and Flexbox wrappers with media queries, ensuring fluid layout on smartphones, tablets, and desktops.

4.5 Font and Character Encoding Consistency

- **Problem:** Raw Unicode emojis in template strings displayed corrupted characters (mojibake) on certain browser configurations.
- **Solution:** Standardized all icons using official **FontAwesome 6 vector icons**, ensuring 100% crisp rendering across all browsers.

5. Individual Contribution

Both group members contributed equally (50% / 50%) across all three phases of the project as summarized below:

Project Phase / Task	Kavindu Kaushalya (ITT/2024/057)	Sahan Sithumina (ITT/2024/032)
Phase 1: Proposal and Wireframes	• Created UI/UX wireframes for Home, Catalog, and Dashboard. • Outlined planned user features.	• Wrote the initial Project Proposal document. • Researched the required technology stack.
Phase 2: Frontend Layout and Design	• Developed Recipe Catalog (<code>recipes.php</code>). • Implemented JavaScript category filtering. • Styled recipe cards and fixed responsive grid.	• Implemented the Home Page hero banner (<code>index.php</code>). • Designed Contact Us frontend layout. • Structured base CSS styles and footer.
Phase 3: PHP and MySQL Backend	• Built User Recipe Dashboard (<code>dashboard.php</code>). • Dynamic Catalog Modal Integration. • Updated dynamic navbar session state.	• Designed the MySQL database (<code>database.sql</code>). • Configured PDO connection (<code>includes/db.php</code>). • Built User Registration and Login with BCRYPT. • Built Contact form database insertion.
Documentation and Testing	• Frontend responsiveness and cross-browser testing. • Final Project Report preparation and formatting.	• Database schema verification in phpMyAdmin. • Prepared SQL export file and test data. • Documented system design and schema.
Total Contribution	50%	50%

5.1 Summary of Individual Tasks

Kavindu Kaushalya (ITT/2024/057): Kaushalya focused on the frontend user interface, client-side JavaScript search and modal functionality, the Recipe Submission Dashboard (`dashboard.php`), responsive CSS layouts, and final report documentation.

Sahan Sithumina (ITT/2024/032): Sithumina focused on designing the MySQL database schema (`database.sql`), establishing the PDO database connection (`db.php`), building User Registration and Login with secure password hashing (`register.php` , `login.php` , `logout.php`), and saving contact inquiries (`contact.php`).

6. Setup Instructions (Running via XAMPP)

To set up and run "Digital Recipe Book" locally on a computer using XAMPP, follow these simple steps:

- **Prerequisites:** Download and install XAMPP (with Apache and MySQL) from <https://www.apachefriends.org/>.
- **Step 1: Put Project in XAMPP htdocs:** Copy the project folder `digital-recipe-book` into `C:\xampp\htdocs\`.
- **Step 2: Start Apache and MySQL:** Open XAMPP Control Panel and click the 'Start' button next to both Apache and MySQL. Make sure both turn green.
- **Step 3: Import the Database:** Open your web browser and go to <http://localhost/phpmyadmin/>.
 - Click on the 'Import' tab at the top.
 - Click 'Choose File' and select `database.sql` from the project folder.
 - Click 'Go' at the bottom. This will automatically create the `recipe_db` database and its 3 tables.
- **Step 4: Check Database Settings:** Check `includes/db.php` in the project. The default settings match standard XAMPP (host: localhost, dbname: recipe_db, user: root, password: empty).
- **Step 5: Open and Test the Website:** Open your browser and visit: <http://localhost/digital-recipe-book/index.php>.
You can now explore recipes, register a new account, log in, submit a recipe via the Dashboard, and send a message through the Contact page.

7. References

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